

DEVELOPMENT PROJECT

SEAT RECLINER BEARING



UNLOCKING
YOUR
ENGINEERING
PUZZLES

A SAINT-GOBAIN BRAND

Background

Making small parts.
Making a big difference.

Seat Recliners can be found in every car

They are a safety critical structural link between the seat cushion frame and the backrest frame

They need to enable backrest angle adjustment whilst playing a key role in load path management, occupant safety, and energy absorption in high severity impacts (crash events).

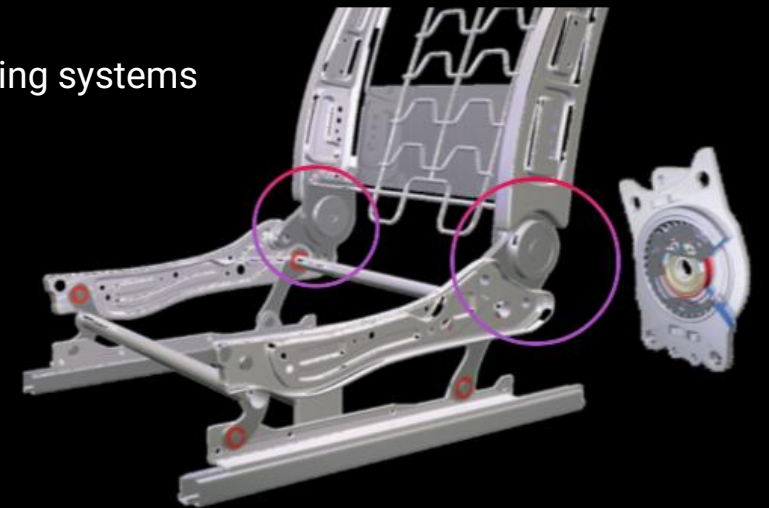
Bearings used are round rings with a closed gap which are greased and covered by a welded lid

The market size is more than 330 million sliding elements worldwide (40 mio EU).

All major TIER-1 suppliers emphasize in their financial reports that they continue to invest in seating systems

Major players are GGB and CSB who offer DU-type solutions for 0.1 – 0.2 €

EQYO currently has < 1 % market share due to PRO and HP parts perceived as too expensive



SAINT-GOBAIN • MOBILITY

EQYO

A SAINT-GOBAIN BRAND
SAINT-GOBAIN

Project Goal

Making small parts.
Making a big difference.

Develop a closed-gap seat recliner bearing for EU OEM seating systems that meets Tier-1 performance requirements at ≤ 0.08 €/pc (var. costs) and enables ≥ 10 % market share in the EU recliner bearing market

SAINT-GOBAIN • MOBILITY

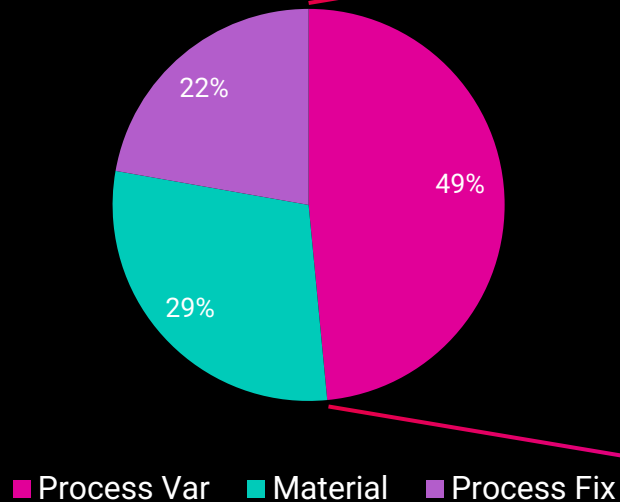


Background - Current Situation

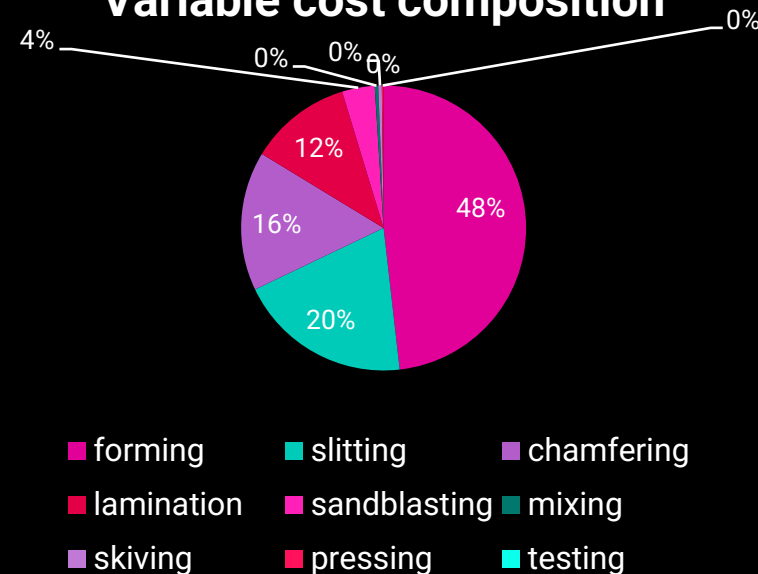
Example: Fisher Seat Recliner Bearing (SAP 123463) made of PRO100E-5007-B

One part costs 17.68 €ct before margin (extracted from SAP)

Total cost composition



Variable cost composition



Possible scenarios for success could be:

- Reduce Material cost by 40 % and forming cost by 50 % with elimination of chamfering & sandblasting (→ 0.080 €ct /pc)
- Reduce Material cost by 30 % and forming cost by 55 % with elimination of chamfering & sandblasting (→ 0.078 €ct /pc)

Project Scope

Conditions to achieve a successful solution

Making small parts.
Making a big difference.

Information from
Market Case

Target variable costs for a bearing $\leq 0,08$ € - requires lower materials costs as well as lower process costs esp. forming

Targeted material thickness of 1.0 mm (later on scalable to 1.25 mm and 1.5 mm)

Bearing must have a closed gap without any bumps (e.g. NO welding or puzzle gap) to avoid tangling

Material must have high load capability (min. 400 N/mm² and Deformation under load ≤ 5 μ m (like PRO))

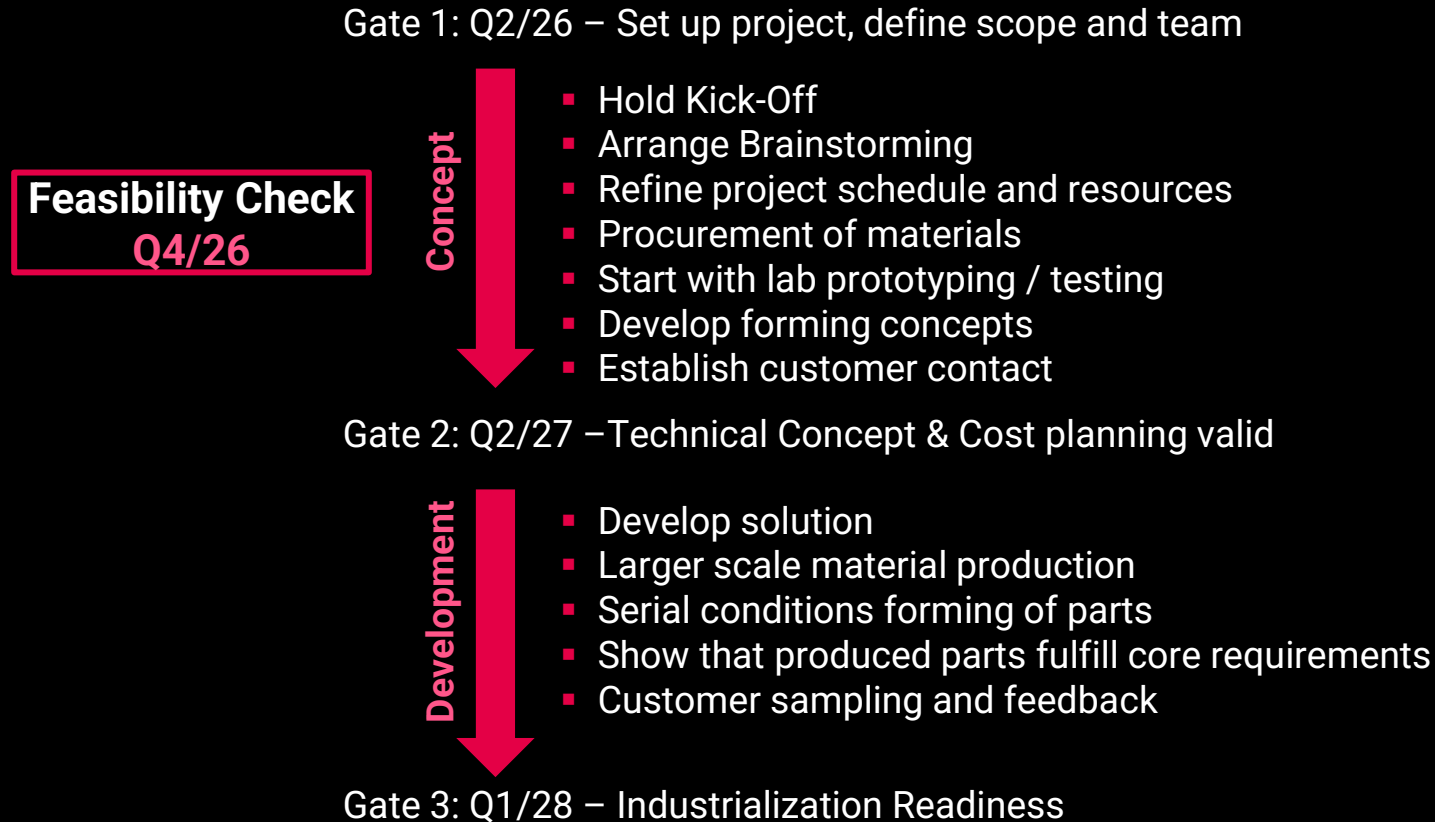
High temperature stability (600 °C < 1 min) to survive welding of lid and possible e-painting

Chamfering capability of top and back edges to ease assembly (min. metal side)

Low COF in greased application ($\sim 0,04 - 0,07$) and high wear resistance (> PROE)

Action program

Making small parts.
Making a big difference.



SAINT-GOBAIN • MOBILITY



Summary

Boundary conditions (fixed):

- Target cost: ≤ 0.08 €/pc
- Timeline: 6 months → feasibility decision
- Core requirements: Load, COF, temperature

Expected outputs after 6 months:

- Validated cost model
- 1–3 feasible technical concepts
- Identified key risks
- Clear Go / No-Go decision

Set-up:

Task structure around 4 main Project Bricks:

- Material strategy - cheaper fillers / structure, reduced thickness vs. stiffness compensation
- Process strategy - enable high-speed forming (→ cost driver!), eliminate secondary steps (chamfering, blasting)
- Design challenge - closed gap without complex joining, reduce tolerances where possible
- System thinking - optimize bearing + grease interaction

Approach & concepts

Methods & experiments

THANK YOU ANY QUESTIONS?

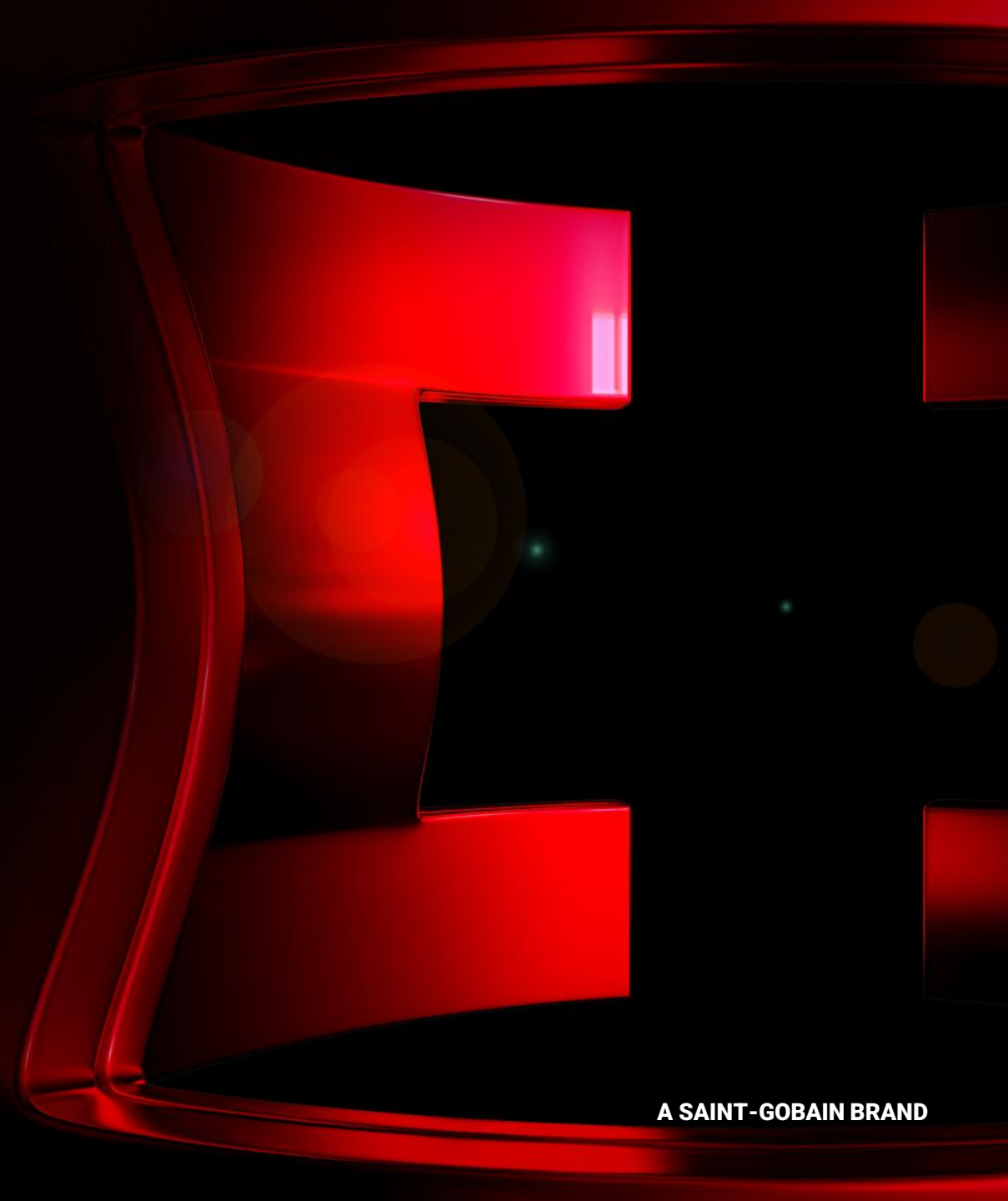
Explore more
EQYO.COM

 youtube.com/@SGEQYO

 linkedin.com/company/eqyo

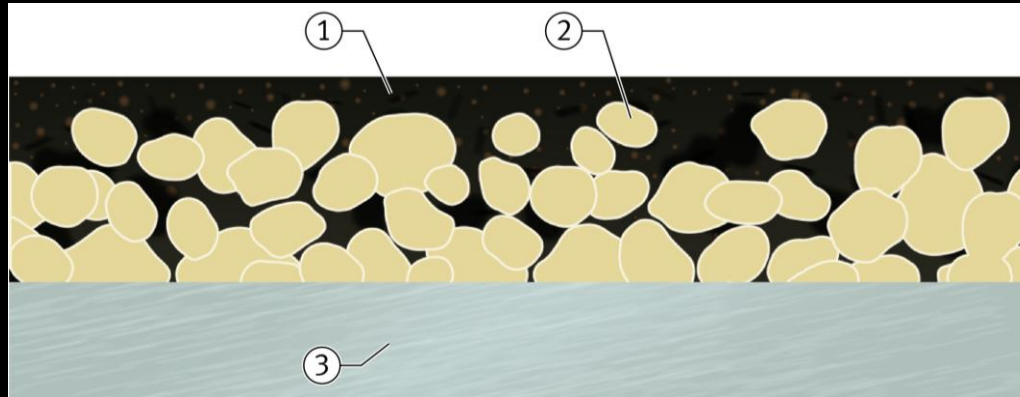
EQYO UNLOCKING
YOUR
ENGINEERING
PUZZLES

A SAINT-GOBAIN BRAND



Why metal-polymer plain bearings work - and how to rethink them

Making small parts.
Making a big difference.



- ① PTFE top layer
- ② Sintered bronze
- ③ Steel backing

Why this works:

- Key design principle: Separation of functions
- Functional breakdown:
 - E.g. Load carrying
 - ...
 - ...

Current challenges:

- PFAS dependency (PTFE)
- Cost structure (multi-layer incl. bronze + sinter process)
- Process complexity

➡ How can we solve these functions in completely different ways?